

PD SMART SUPPLY

REV A DESIGN DEFINITION AND BUILD RECORD

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Date	August 2026	Status	Design complete

1. PURPOSE AND SCOPE

PD Smart Supply is a compact USB-C Power Delivery powered programmable supply. It requests a selected PD voltage, protects and monitors the power path, and provides two parallel protected outputs. An STM32 microcontroller manages voltage selection, supervision, display and user input. Rev A is intended as a functional engineering prototype.

2. PRINCIPAL SPECIFICATIONS

Item	Specification
Input	USB-C PD through a USB4105-GF-A receptacle
Selectable output	5 V, 9 V, 12 V and 20 V; 28 V intentionally disabled
Maximum output	20 V at 3 A, approximately 60 W
Output connectors	Two parallel outputs sharing a combined 3 A rating
Controller	STM32G071CBT6
Display	HS96L03W2C03, 0.96 inch, 128 x 64, I2C OLED
Controls	UP, DOWN, SELECT and RESET tactile buttons
Board	70 x 50 mm, four layers, nominal 1.6 mm FR-4

3. SYSTEM ARCHITECTURE

Main power path: USB-C input -> CH224A PD controller -> VBUS_PD -> TPS259470L eFuse -> EFUSE_OUT -> OUT1 and OUT2.

Auxiliary power path: VBUS_PD -> TPS560430 buck converter -> 3.3 V logic rail -> MCU, OLED and control circuitry.

The two output connectors are electrically parallel. Their total permitted load is 3 A, not 3 A per connector.

USB-C PD control

The CH224A negotiates the requested source voltage. CC1 and CC2 connect directly to the controller. USB D+ and D- are routed as a short pair for charging-protocol support. Three MCU control lines select the requested voltage. Firmware must not generate the CH224A state that requests 28 V because the Rev A protection design is limited to 20 V.

4. POWER PROTECTION AND REGULATION

Protected output

The TPS259470L eFuse provides inrush control, current limiting, monitoring and MCU-controlled shutdown. A TVS2200 protects VBUS_PD, and the output clamp diode limits negative transients.

Parameter	Rev A value
User output rating	3 A combined across OUT1 and OUT2
Nominal current limit	Approximately 3.76 A using 887 ohm RILM
Overvoltage threshold	Approximately 22.1 V
Output slew rate	Approximately 0.91 V/ms; about 22 ms to 20 V
Fail-safe state	eFuse defaults off while the MCU is in reset

3.3 V supply

A TPS560430 buck converter derives the 3.3 V rail from VBUS_PD. The stage uses a 10 uH inductor, 22 uF output capacitor and 51 k / 22.1 k feedback divider. The switching loop and feedback network are kept compact and separated from analog sensing.

5. MEASUREMENT AND CONTROL

Signal	Purpose or scaling	MCU
VOUT_SENSE	VOUT / 10.98; approximately 1.82 V at 20 V	PA0 ADC
IOUT_SENSE	Approximately 0.161 V/A; 0.484 V at 3 A	PA1 ADC
EFUSE_nFLT	Active-low fault indication	PB15
EFUSE_READY	Valid input and completed inrush indication	PB14
EFUSE_OFF	High disables the protected output	PB13

User and service interfaces

Interface	Implementation
Navigation	UP, DOWN and SELECT buttons with 10 k pull-ups
Reset	Dedicated pushbutton connected to NRST
Display	128 x 64 I2C OLED operating from 3.3 V
Status LED	Driven from CH224A power-good logic through a 2N7002
Debug	Four-pin SWD header: SWDIO, SWCLK, NRST and GND
Test points	VBUS_PD, 3V3, VOUT, VOUT_SENSE, IOUT_SENSE and GND

6. FIRMWARE REQUIREMENTS

Firmware shall permit only 5 V, 9 V, 12 V and 20 V selections; keep the eFuse disabled until startup checks pass; wait approximately 5 ms after voltage changes before accepting VOUT_SENSE; and use EFUSE_nFLT for fault response. EFUSE_READY alone is not a complete fault indicator.

7. PCB IMPLEMENTATION

Item	Rev A definition
Fabricator target	JLCPCB stackup JLC04161H-7628
Dimensions	70 x 50 mm
Layer order	F.Cu signal/power - In1.Cu ground - In2.Cu power - B.Cu signal/ground
Finished thickness	KiCad model 1.6062 mm; nominal order thickness 1.6 mm
Copper	1 oz outer layers; 0.5 oz inner layers
Finish	HAL SnPb, green solder mask, white silkscreen
Assembly population	63 references in the verified BOM and placement file

Layout

The high-current path runs directly from USB-C through protection and the eFuse to the output connectors. In1 is the continuous ground reference. The buck switching node is compact, analog sense traces are kept away from the inductor, and ground stitching is concentrated around the USB, protection, buck, MCU and connector regions.

8. MANUFACTURING STATUS

Check	Result
Routing	Complete; final verification found zero unconnected items
Fabrication package	Four copper layers, masks, paste, silkscreen, profile, PTH and NPTH drills verified
Assembly files	BOM and CPL match at 63 references; all BOM rows contain LCSC numbers
Accepted exception	USB clearance of 0.2462 mm against the project rule of 0.2500 mm
Nonfunctional warnings	Known footprint silkscreen clipping and mechanical/test-point parity warnings

Procurement note

CH224A, JLCPCB/LCSC C42459160, was out of stock during the August 2026 quotation and was excluded from assembly. The exact ESSOP-10 device must be sourced and reflow-soldered before complete USB-PD operation can be tested.

9. INITIAL BRING-UP

Step	Required check
1	Inspect assembly, component polarity and resistance to ground
2	Apply current-limited 5 V and verify VBUS_PD and 3.3 V
3	Program the MCU and verify reset, buttons, OLED and LED
4	Fit and verify the CH224A if it was not assembled
5	Test 5 V, 9 V, 12 V and 20 V negotiation without load
6	Increase load gradually while checking regulation, sensing, protection and temperature

10. CONCLUSION

Rev A is design-complete and ready for prototype manufacture. Physical bring-up will validate electrical behavior, thermal performance, USB-PD interoperability, measurement accuracy, firmware control and enclosure integration before any Rev B changes are considered.